

Cybersecurity Incident Report: Travel Agency

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is that the server is experiencing a DoS/SYN flood attack. High-volume malicious traffic sent by the attacker and the IP address is flooding the website server with repeated SYN connection requests. The server's resources are overwhelmed, and while trying to handle the flood attacks, it is failing to respond to legitimate users.

The logs show that a severe TCP SYN flood attack occurred targeting the server at 192.0.2.1 on Port 443. The attacker sent a continuous stream of SYN packets to overwhelm the server's connection queue.

This event could be: The event captured here is a SYN Flood where an attacker sends a succession of SYN requests to a target system in an attempt to consume enough server resources to make the system unresponsive to legitimate traffic.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The first step is SYN. The visitor's device initiates a connection by sending a packet with SYN. This tells the server that the client wants to establish a connection and establishes the initial sequence number.
2. The website server receives the request and responds with a SYN-ACK, which is a SYN acknowledgement. This acknowledges that it received the client's initial request and send it own SYN signal with its sequence number.
3. The visitor's device receives the server's response and sends a final ACK packet back. This acknowledges receipt of the server's SYN-ACK packet and completes the connection setup. Once these three steps are complete, the TCP connection is fully established. Immediately following this handshake, the actual website data transfer begins.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server drops or forcefully resets normal connections, resulting in severe connection failures and resets. Multiple gateway timeouts cause the server to ultimately

become entirely unresponsive to real users. Multiple legitimate source IPs were disrupted during this window.

Explain what the logs indicate and how that affects the server: The network logs reveal a severe TCP SYN flood attack that exhausts the server's resources, causing it to become unresponsive and drop legitimate user connections.